

A genetic approach for simulating persistence of reintroduced tree species populations in restored forests

Patricia S. Sujii^{a,1,*}, Micael E. Nagai^b, Maria I. Zucchi^c, Pedro H.S. Brancalion^d, Patrick M.A. James^e

^a Genetics and Molecular Biology Graduate Program, Biology Institute, Unicamp, Campinas, 13083-970, Brazil

^b Ecology Graduate Program, Biology Institute, Unicamp, Campinas, 13083-970, Brazil

^c Agência Paulista de Tecnologia dos Agronegócios, Pólo Regional Centro Sul, Piracicaba, 13400-970, Brazil

^d Department of Forest Sciences, “Luiz de Queiroz” College of Agriculture, University of São Paulo, Piracicaba, 13418-900, Brazil

^e Département de Sciences Biologiques, Université de Montréal, Montréal, QC, H3C 3J7, Canada

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ABSTRACT

Tree populations in regions undergoing restoration are generally made up of few individuals, isolated from neighboring populations, and are found within a matrix of inhospitable human-modified landscapes. Resulting negative genetic consequences such as inbreeding depression and genetic drift require mitigation strategies to maintain sufficient genetic diversity in restoration areas. Such strategies often involve seed sampling from many source trees with different provenances. However, the efficacy of these approaches has not been validated. We present an individual-based spatial simulation model to evaluate the effects of: 1) differing levels of initial genetic diversity; and 2) different area sizes on short (tens of years) and mid-term (hundreds of years) restored population viability. We demonstrate this approach and the use of our model with case study of *Centrolebium tomentosum*, a tropical tree species widely used in restoration projects in the Atlantic Forest of Brazil. Our model and analysis framework can be applied in studies of tree species with different characteristics, from tropical and temperate forests, to assess population persistence in restoration sites as a function of genetic diversity and population size. This knowledge can support planning of both restoration projects and management actions, increasing the probability of restoration success while also reducing associated costs.

1. Introduction

Ecological restoration endeavors to recover and improve ecological processes to attain functional and resilient communities that can adapt to changing conditions (Alexander et al., 2011). Restoration is now acknowledged as a global environmental priority and many international initiatives are aiming to restore millions of hectares around the world (Suding et al., 2015; Holl, 2017). Broad scale re-conversion of pasturelands and agricultural fields to native forest ecosystems is seen as necessary to mitigate future extinctions in regions with high levels of habitat loss and fragmentation (Banks-Leite et al., 2014; Newmark et al., 2017). The goal of such restoration is to support conservation, re-establish plant populations with sufficient genetic diversity to persist overtime, and to increase landscape connectivity to facilitate plant and animal gene flow in human-modified landscapes (Chazdon et al., 2015; Brancalion et al., 2018).

Despite the myriad of benefits that restoration may provide, the probability of population persistence following forest restoration planting is uncertain, and can be affected by multiple factors including micro-site limitations for recruitment (Bertacchi et al., 2016), competition with invasive species (D’Antonio and Meyerson, 2002), pollination and dispersal limitation (Dixon, 2009), climate change (Harris et al., 2006), and genetic-mediated processes, such as inbreeding depression (Thomas et al., 2014). Improving our understanding of the conditions under which persistence is more likely is critical to successful restoration and effective conservation (Suding et al., 2015).

The use of computer models is one potential solution for this problem (Epperson et al., 2010). The process through which data and models of populations are evaluated to estimate likelihoods of population persistence over an arbitrary amount of time is called population viability analysis (PVA; Boyce, 1992). Some research groups developed

* Corresponding author.

E-mail addresses: patricia.suji@udf.edu.br (P.S. Sujii), micaelnagai@yahoo.com.br (M.E. Nagai), mizucchi@apta.sp.gov.br (M.I. Zucchi), pedrob@usp.br (P.H.S. Brancalion), patrick.ma.james@umontreal.ca (P.M.A. James).

¹ Present address: Health School, Centro Universitário do Distrito Federal, Brasília, DF, 70390-045, Brazil.

forest models with varied applications, such as to predict succession dynamics of forest communities (Pacala et al., 1996), understand individual tree growth (Hauhs et al., 1995) and community dynamics (Bugmann, 2001), and evaluate the interaction of trees with environmental factors (Seidl et al., 2012). Also, an increasing number of forest landscape simulation models are available, as reviewed by Scheller and Mladenoff (2007). However, there is an unfortunate lack of simulation programs that combine population dynamics and spatial and temporal genetic variation of tree species, which have long life spans, and overlapping generations. Also, most PVAs assessments ignore or inadequately model genetic factors as the effect of inbreeding depression on total fitness (Frankham et al., 2014).

Mid- and long-term persistence of restored tree populations depends on the capacity of species and populations to evolve in response to environmental changes. This capacity in turn depends on intra-specific population levels of genetic diversity (Booy et al., 2000). Populations with reduced genetic diversity tend to show reduced fitness, limited potential for adaptation (Reed and Frankham, 2003), and have a higher probability of extinction due to diseases or environmental stochasticity (Mills, 2012). Thus, using seeds with high genetic diversity in restoration projects is one of the main strategies for effective restoration and successful conservation in the face of environmental change (Zucchi et al., 2018). However, our understanding of how to use genetic data in restoration projects remains incomplete (Thomas et al., 2014; Basey et al., 2015).

Efforts to include genetic diversity in forest restoration often involve the selection of specific source populations and a diversity of trees from which seeds are collected for sapling production (mother trees) for seed harvesting (Bozzano et al., 2014; Basey et al., 2015). Restoration practitioners typically aim to ensure the long-term persistence of re-introduced populations by using seeds from local populations, which increases the probability that individuals will be adapted to local environmental conditions. Practitioners also aim to maximize the genetic diversity of the founding population through the selection of many different mother trees (Hufford and Mazer, 2003; McKay et al., 2005).

Restoration projects that aim to increase genetic diversity and local adaptation in seedling populations nonetheless face significant challenges when implemented at small geographic scales, because small areas comprise small populations. Because of their relatively low effective population size, these populations are at risk of loss of genetic diversity similar to populations negatively affected by habitat loss and fragmentation (Sezen et al., 2007; Chazdon, 2014). The loss of genetic diversity can be observed as a reduction in heterozygosity and allelic diversity through time. The smaller the population, the faster the manifestation of the negative effects of drift on genetic diversity is expected (Mills, 2012). The loss of genetic diversity in these populations due to inbreeding depression further increases the risk of local extinction (Frankham, 2005) and, consequently, restoration failure.

To address this issue, researchers and practitioners have provided different recommendations for the minimum population size necessary to avoid the negative effects of genetic drift and inbreeding depression in the context of restoration in both the short-term (5 generations) and long-term (> 100 years). It has been suggested as a general rule that seeds should be sampled from at least 30 (Vencovsky, 1986) or 50 trees (Basey et al., 2015) to recover sufficient genetic diversity in tree populations with an effective population size (N_e) ≥ 100 . However, there is a lack of studies validating or testing the efficacy and feasibility of these recommendations due to constraints in both ecological (limited number of mother trees of low density species in fragmented landscapes) and financial terms (time and resources spent to find numerous mother trees and collect their seeds) (Brancalion et al., 2012). For example, in tropical forest restoration efforts, where over 100 native tree species have been reintroduced in restoration plantations (Rodrigues et al., 2011), reaching the targeted number of mother trees per species (i.e., 30–50) can be difficult. Also, as the current recommendations for

forest restoration is to plant 20 in/ha, in small restoration areas (< 5 ha), up to two or three seedling from each mother tree would be planted if the general rule is followed. Improved evaluation of the probability of success of restoration approaches that vary in the number of mother trees used and the effective population size targeted is needed to effectively deploy limited restoration resources.

In this study we present a novel individual-based model to simulate the spatio-temporal genetic and population dynamics of trees in restoration plantations in response to different restoration strategies and over long time scales. Using this model, we specifically examine how different levels of initial genetic diversity and population size influence long-term restoration success, defined as short and mid-term population persistence. Currently, there are few programs that simulate the population dynamics of tree species, which have long life spans, and overlapping generations, and that also model spatial and temporal genetic variation. Also, most of these models ignore or inadequately capture genetic factors such as the effect of inbreeding depression on total fitness (Frankham et al., 2014). In our expository treatment of our newly developed model, we present a case study application based on restoration of a tropical tree in the Atlantic Forest of Brazil.

2. The model

The description of our model follows the **ODD** (Overview, Design concepts, Detail) protocol of Grimm et al. (2006, 2010).

This model was designed to investigate the influence of initial genetic diversity and area size on population persistence. Initial genetic diversity within a restoration plantation is represented by the minimum number of mother trees from which seeds were collected. In addition to initial area size, this is one of the parameters over which management agencies have the greatest control when designing and implementing restoration strategies. In this context, we set out to address the following two specific questions: 1) How does area size and initial genetic diversity affect the loss of genetic diversity of restored tree populations? And, 2) what is the minimum number of mother trees from which seeds should be collected for a restoration forest plantation to ensure sustainable levels of genetic diversity in the restored population?

2.1. Entities, state variables, and scales

The entities of this model are individual trees. Each individual is characterized by the following three state variables: 1) spatial location; 2) age; and 3) multi-locus genotype. Spatial location describes the geographic (x and y) coordinates on a two dimensional lattice. Tree age varies from zero to a maximum species age and increases by one unit (i.e., year) per time step. Simulations begin with all individuals at three years old, as in a recent plantation. Seeds have age of zero, and germinated seedlings have age of one. Individuals are considered juveniles before they reach reproductive age and adults are reproductive until the end of life with the same reproduction rate (Fig. S1). Each genotype is composed of multiple independent multi-allelic co-dominant loci with no linkage disequilibrium (Fig. S2). Alleles are represented by a pair of integer numbers each of which can take a value between one to the maximum number of alleles initially defined.

The time step in the model is one year. The spatial extent of the simulated area consists of a single isolated rectangular forest fragment (patch) undergoing restoration. The area shape, always a narrow rectangle, was designed to simulate narrow riparian forests, where there is no light limitation, and that have received special attention over the past 30 years, due to their importance to river protection (Rodrigues et al., 2009). These forests patches are typically long and narrow and are intensively affected by edge effects (Didham and Lawton, 1999; Davies-Colley et al., 2000) and tend to retain the characteristics of early-successional systems as they age (Tabarelli et al., 2008).

2.2. Process overview

The model starts by defining initial conditions, including age, spatial location, and genotypes of individual trees in the initial planted population. The following model parameters are also specified: simulation duration (*time*); number of simulation repetitions (*rep*); number of loci in the input dataset; lattice dimensions (*xDim*, *yDim*); pollen (*dPollen*) and seed (*dSeed*) dispersal distances; number of seeds produced by each tree (*avSeed*); maximum age an individual can reach (*maxAge*); at what age the individual becomes an adult (*adultAge*); maximum number of pollen donors for each offspring (*maxFathers*); germination probability (*germ*); selection factor (*selection*); and the time steps when output files should be saved (*output_years*).

Each cycle starts with the determination of each individual's survival rate, as a function of age. This model consider hermaphrodite tree species, so all individuals can be either mother or father of an offspring, and self-fertilization is possible. The reproduction step starts by defining for each mature tree (i.e., mother), which mature trees (i.e., potential fathers) are present within the pollen dispersal range. If the number of potential mates is fewer than the *maxFathers*, all mates contribute to pollination. Otherwise, a number of potential mates equivalent to *maxFathers* are randomly selected to be pollen donors. By default, potential mates as sampled with a uniform probability distribution, but other probability distributions may be implemented. Alleles are inherited following Mendelian inheritance.

Seed location is defined by the seed dispersal range and a randomly selected dispersal direction. If a seed falls at the same location of an existing adult or out of the study area, it does not survive. Otherwise, each seed has the predefined initial probability of germination, which can be decreased as a function of its inbreeding coefficient (Richards, 2000). In species that produce a large number of seeds from which only a few germinate and survive to juvenile phase, genetic drift tends to have a stronger effect than selection. Thus, selection on germination and survival is not strong enough to purge deleterious alleles efficiently (Keller and Walter, 2002). In these situations, inbreeding depression can affect seed and seedling viability (Naito et al., 2005), germination success (Richards, 2000), and survival rates (Ishida, 2006). The assumptions for estimating the inbreeding effects on fitness in this model are: (1) an increase in inbreeding by 10% leads to a reduction in fitness components of 5–10% (Frankel and Soulé, 1981); (2) a large number of loci are affected by inbreeding depression (Ayroles et al., 2009; Chaves et al., 2011); and (3) a sample of loci allows us to infer the inbreeding coefficient of a population (Chaves et al., 2011).

When a new seed successfully germinates it is added to the tracking list of active individuals with an age of one. Probability of survival to the next year for both juveniles and reproductive individuals is assessed in each time step as an increasing function of tree age (Fig. S1). Dead individuals are removed from the tracking list and hence the population.

At the end of the simulation, a table summarizing information on the location, genotype, age, and survival probability for all living individuals is produced. Genotypic output data are converted to the FSTAT (Goudet, 1995) format and genetic summary statistics are calculated for all individuals with age ≥ 1 , using the software *Hierfstat* (Goudet, 2005). Summary statistics calculated include: number of individuals, total number of alleles, allelic richness, observed and expected heterozygosity under Hardy-Weinberg Equilibrium (HWE), and the inbreeding coefficient with confidence interval.

2.3. Design concepts

2.3.1. Basic concepts

The basic principles of this model are that the population is isolated from others (no migration), mutation rates are negligible, individuals only mate with others within the pollen dispersal range (i.e., the

population is not panmictic), and germination probability is influenced by the inbreeding coefficient of each seed. Mortality in all other life-stages is age dependent (Fig. S1).

2.3.2. Emergence

Population size is the total number of individuals with minimum age of 1 year old and is a consequence of mating rules (see below), survival probabilities, and the amount of space available. This means that population size is an emergent property of the simulation. The frequency and number of alleles vary as a result of genetic drift, which is random although influenced by the assumptions of dispersal and overall model spatial scale. Thus allele frequencies and their spatial distribution are an emergent property.

2.3.3. Stochasticity

Randomness in the model operates through mating and survival. Mates are sampled at random from the adults available within the pollen dispersal range. Other than distance from the mother tree, there are no other rules for mating preference. Seed genotypes are the result of random sampling of the parents' genotypes. Each seed's location is defined in the seed dispersal range from the mother tree, according to the rules described in the seed dispersal sub-model, which includes randomness in dispersal direction. Death is probabilistic and depends on the age of the individual (Fig. S1).

2.4. Initialization

At the beginning of each simulation, the parameters of the initial population planted in the area must be defined. The initial population consists of saplings, all the same age, that may descend from different number of mother trees. Their genotypes are drawn depending on the number of mother trees from which they descended and may vary from a pool of individuals in HWE to a full sibling pool. The saplings can be planted uniformly, randomly, or aggregated in one or more clusters, as defined by the user.

2.5. Input

The input dataset consists of a tab-delimited data table comprised of desired initial individual locations (x, y), genotypes, and age (Fig. S2). There is no limitation on number of individuals and the size of the restoration area, except for those imposed by the memory limitations of each user's computer system.

2.6. Sub-models

The model contains four sub-models: 1) reproduction; 2) seed dispersal; 3) seed germination; and 4) age-specific mortality.

2.6.1. Reproduction

This sub-model simulates sexual reproduction among individuals in the pollen dispersal distance range. The model does not simulate asexual reproduction or gene flow from other, non-explicitly simulated populations. Reproduction occurs once per annual time-step. All adult individuals are considered to be reproductive and all have the same potential for pollen and seed production. We assume that there is no pollen limitation. Pollen donors are randomly sampled from the pool of individuals in the pollen dispersal distance range. We assumed Mendelian inheritance of alleles, as the model's loci represent independent nuclear genetic marker loci (i.e., no linkage). Selfing was allowed in the simulations, as the mother tree is included in the pollen donors.

2.6.2. Seed dispersal

We assumed isotropy (no preferential direction for dispersal), so the

Table 1
Parameters and values of the model.

Parameter	Value/ Distribution	Reference
Reproduction		
Sexual maturity age	age = 10 years	Brancalion (communication)
Maximum pollen dispersal distance	$D_{\text{pollen}} \leq 1,000$ m	Pasquet et al., 2008; Hagen et al., 2011
Pollen donors	$n \leq 20$	Sujii, 2015.
Seed dispersal		
Maximum seed dispersal distance	$D_{\text{seed}} \leq 100$ m	Sujii, 2015.
Seed dispersal function	Weibull distribution ($k = 0.6942$; $\lambda = 81.4807$)	Sujii, 2015.
Germination		
Germination probability function	$F_{IS} \leq 0, g = 0.7$ $F_{IS} > 0, g = 0.7 \cdot (1 - 0.5 \cdot F_{IS})$	Carvalho, 2005; Frankel and Soulé, 1981; Richards, 2000
Mortality		
Maximum age	age = 89 years	Laurance et al., 2004; Silva, 2013
Survival rate	age = 1, sRate = 0.1 $1 > \text{age} \leq 5$ years, sRate = $0.2 \cdot \text{age} - 0.3$ age > 5 years, sRate = Weibull function ($k = 0.7631$, $\lambda = 7.8216$) age > 89, sRate = 0	

seed location is defined by the dispersal distance function and a random value from one to 2π radians for the direction. All seeds produced in a year germinate or die; we do not consider soil seed bank formation.

2.6.3. Seed germination

We modelled the probability of seed germination as a function of inbreeding. Thus, in this sub-model, the probability of germination is estimated using all neutral loci, wherein if the inbreeding coefficient (F_{IS}) is greater than zero, an increase in inbreeding leads to a reduction of germination rate in the scale defined by the selection factor ($gRate = g \cdot (1 - \text{selection} \cdot F_{IS})$), where g is the probability of germination with no inbreeding. So, a selection factor of 0.5 means that an increase in inbreeding by 10% leads to a reduction of 5% in germination rate (Frankel and Soulé, 1981). If inbreeding coefficient is zero or negative, it has no effect on germination rate.

2.6.4. Survival

The probability of mortality increased as a function of age (Table 1) up until the maximum age for that species, at which point all individuals die. To determine whether an individual dies at each time step, a random number is sampled from a uniform distribution between zero and one. If the number is larger than the survival probability (sRate) associated with the individual's age, then the individual dies.

2.7. Model development

We wrote the program script in the scientific computing language R (R Core Team, 2015). General model schematic is described in Fig. 1 and the program script can be found in supplementary material and also at GitHub (https://github.com/sujiips/Restoration_PVA.git).

2.8. Sensitivity analysis

Sensitivity analyses were performed to quantify the relative importance of each model process on simulation results and to assess how uncertainty in parameter values affects simulation dynamics. All simulations for sensitivity analysis used a restoration area of five hectares with a presumed seed source of a large population and using reference parameter values from *Centrolobium tomentosum* (Fabaceae), a tree species from an intermediate stage of ecological succession from a tropical forest. Each simulation was run for 500 time steps and each scenario for a unique set of parameters was replicated 250 times. This number of replicates was found to capture sufficient variation in system dynamics (i.e., $CV < 0.05$; Fig. S3).

We performed a local sensitivity analysis (Railsback and Grimm, 2011), which involved varying parameter values one at a time. Parameter values were varied between +5% and - 5% (Railsback and

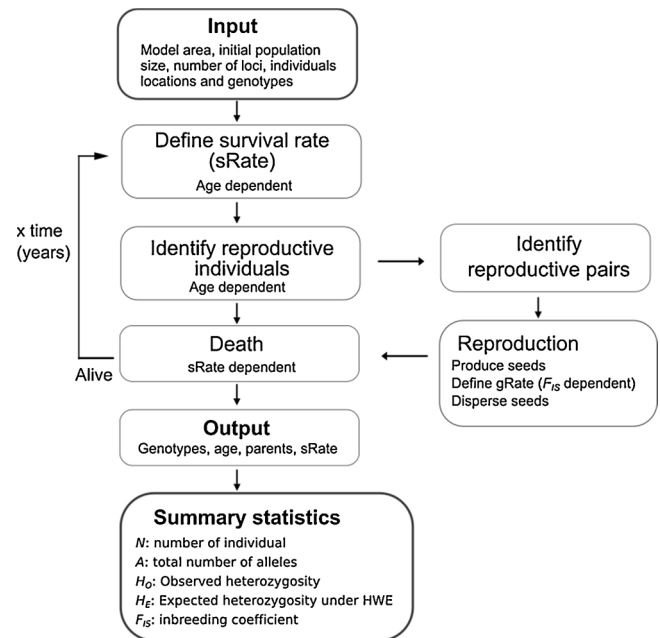


Fig. 1. Model flow chart.

Grimm, 2011), except for some parameters (e.g., adult age and maximum pollen dispersal distance), for which a 5% variation was too small to produce noticeable response. In these situations, a higher percentage of variation was selected (Table 2).

3. Model application

We demonstrate application of our model using a simulated restoration experiment using *C. tomentosum* (Fabaceae) in the tropical Atlantic Forest of Brazil. This species was selected because it is widely used in restoration projects in Brazil and because demographic and genetic data are readily available. *C. tomentosum* is frequently used for forest restoration because it is a common gap-colonizing species, reproduces once a year, grows quickly, produce high-quality timber, and has symbiotic associations with nitrogen fixation microorganisms which facilitate its rapid colonization (Carvalho, 2005; Pagano, 2008). This species has a wide distribution over the Atlantic Forest and some parts of Cerrado, two of the most threatened global hotspots for biodiversity conservation. We investigated the consequences of varying three experimental factors on genetic diversity and short and mid-terms population viability in forest patches that have undergone restoration planting. These factors were: 1) initial population size; 2) size of the

Table 2

Values for number of alleles (A), inbreeding coefficient (F_{IS}) and proportion of extinct populations estimates in the local sensitivity analysis and the sensitivity coefficient (S).

Variable	Value	Variation	A		F_{IS}		Proportion of extinct populations	
			value	S	value	S	value	S
Reference condition	*		1.83		0.396		0.480	
Adult age	5 yrs	–50%	2.52	1.37	0.204	–0.383	0.388	–0.184
Adult age	15 yrs	+ 50%	1.66	–0.35	0.473	0.155	0.676	0.392
Pollen dispersal	500 m	–50%	1.79	–0.08	0.458	0.125	0.560	0.160
Pollen dispersal	1500 m	+ 50%	2.00	0.33	0.354	–0.084	0.532	0.104
Maximum number of pollen donors	15	–25%	1.88	0.18	0.362	–0.136	0.484	0.016
Maximum number of pollen donors	25	+ 25%	1.95	0.48	0.347	–0.194	0.528	0.192
Seed dispersal	50 m	–50%	2.98	2.30	0.097	–0.597	0.136	–0.688
Seed dispersal	150 m	+ 50%	1.55	–0.56	0.531	0.271	0.676	0.392
Germination rate	0.6	–14%	1.50	–2.37	0.592	1.376	0.752	1.904
Germination rate	0.8	+ 14%	2.52	4.80	0.168	–1.594	0.260	–1.540
Selection pressure value	0.475	–5%	1.93	2.01	0.358	–0.751	0.448	–0.640
Selection pressure value	0.525	+ 5%	2.01	3.62	0.321	–1.502	0.520	0.800
Maximum age	85 yrs	–5%	1.71	–2.81	0.424	0.641	0.592	2.492
Maximum age	93 yrs	+ 5%	2.14	6.86	0.272	–2.758	0.440	–0.890
Average number of seeds per tree	74	–5%	2.04	3.58	0.346	–0.878	0.436	–0.770
Average number of seeds per tree	67	+ 5%	1.595	–5.549	0.512	2.723	0.576	2.240

Reference values: Adult age = 10 years; Pollen dispersal = 1000 m, Maximum number of pollen donors = 20, Seed dispersal = 100 m, Germination rate = 0.7, Selection pressure = 0.5, Maximum age = 89 years, Average number of seeds per tree = 70.

restoration area, and 3) initial genetic composition (number of alleles and genotypic frequencies). Initial genetic composition was determined by the number of mother trees from which seeds were collected. We estimated mean number of alleles, effective number of alleles, allelic richness, inbreeding coefficient and probability of extinction to evaluate the effects of the experimental factors on each population.

For the purposes of this experiment, simulated forest patches represent riparian areas (long and narrow), with a fixed 30 m width, as are many restoration plantations (Rodrigues et al., 2009). Simulations were run over 500 years and output files were saved at 30, 50, 100, 250 and 500 years. We simulated patch areas of 5 ha, 10 ha, and 20 ha. Initial population size varied with patch area because the current recommendations for forest restoration is to plant 20 in./ha.

The tree spatial distribution in the forest patch, the mean number of alleles per locus, the inbreeding coefficient, and the seed dispersal pattern used to produce the initial population were based on data obtained from real populations in well-established (27 and 57 years old) restoration plantations (Sujii, 2015; Sujii et al., 2017). To produce the input datasets, we first created a simulated a large initial population ($N = 1000$), with all loci in Hardy-Weinberg Equilibrium (HWE), and the same number of loci and alleles per locus that would be used in the simulations (10 loci, 5 alleles per locus). Initial populations were composed of seedlings all of the same age, as in a restoration plantation. Seedlings in the input dataset were the result of simulated reproduction events between seed sources and pollen donors from the original population. Therefore, the input dataset represents a sample of seeds collected from a large population, as is typically the case in applied restoration projects.

The genetic composition of each initial population was determined by the number of seed sources and pollen donors. In some initial populations, seedlings were all the result of self-fertilization, as the offspring of one isolated tree. In other initial populations, different number of seed sources and pollen donors contributed with alleles for the offspring (seed sources = 1, 5, 10 or 30 individuals; pollen donors = 5, 10 or 20 individuals). Initial populations created with larger number of seed sources and pollen donors had higher initial genetic diversity. For simulations with saplings from one mother tree and five pollen donors, we sampled one individual from the original population to be the mother and five individuals to be the pollen donors. This mother was the source of the first allele of each locus of each sapling used in the plantation. The second allele of each locus was randomly

sampled from the five pollen donors. For the simulations with saplings from more than one mother tree, we applied the same procedure as for one mother, but using additional sampled individuals as mothers.

The assumed maximum tree age is 89 years based on mean growth rate functions for both canopy and emergent trees (Laurance et al., 2004). Growth rates for *C. tomentosum* was estimated using diameter measures of trees from nine restoration areas with different known ages (6–96 years; Silva, 2013). The probability of surviving to the next time step was estimated using a demographic study of adult individuals from a natural remnant forest fragment (Barreto, 2009; Silva, 2013) and follows a Weibull distribution. As there were no available data for seedlings and sapling survival rates, a linear function was used such that for each year of increase in age between one and six, survival rate increases by 0.2. After the age of six we use empirical data on survival (Barreto, 2009; Silva, 2013; Fig. S1).

As *C. tomentosum* is a hermaphrodite species and all individuals can be either mother or father of an offspring, self-fertilization is also possible. The main pollinators are large bees with long distance flight capacity. The maximum distance of pollen flow (1000 m) was based on information on flight patterns and mean maximum flight distances of large bees, which are the main pollinators of *C. tomentosum* (Pasquet et al., 2008; Hagen et al., 2011). The pollen donors are randomly sampled from the pool of individuals in the pollen dispersal distance range, following the pollinators' foraging behaviour (Keasar et al., 1996). The species has a mixed mating system, and the maximum number of pollen donors for each mother tree was selected to be 20 trees.

C. tomentosum has one or two seeds enclosed in a large (~ 9 g each) thorny samara (winged fruit), that does not remain in permanent seed banks and are dispersed by wind for short distances (at most 100 m away from the mother tree, following a Weibull distribution), so most seeds fall under the canopy of the mother tree (Cavalho 2005; Aidar and Joly, 2003; Sujii, 2015). The germination probability is 0.7 (Carvalho, 2005). A summary of all parameter values can be found in Table 1.

4. Results

4.1. Sensitivity analysis

We found that our model is not overly sensitive to any of the parameters tested given the range of values we examined (Table 2).

Table 3

ANOVA table summarizing the effects of and interactions between initial genetic diversity and patch area on the total number of alleles and inbreeding coefficient in the population.

Factor	Df	total number of alleles			inbreeding coefficient		
		MS	F value	Pr(> F)	MS	F value	Pr(> F)
mothers	1	193.7	150.356	0.000**	1.001	57.069	0.000**
fathers	1	864.6	671.093	0.000**	3.681	209.769	0.000**
area	1	1696.7	1316.988	0.000**	0.232	13.203	0.000**
mothers: fathers	1	72.5	56.274	0.000**	1.376	78.432	0.000**
mothers: area	1	168.1	130.517	0.000**	0.275	2.089	0.000**
fathers: area	1	76.9	59.653	0.000**	0.082	4.654	0.031*
mothers: fathers: area	1	5.8	4.476	0.0344*	115	6.533	0.0106*
Residuals	7492	1.3			0.018		

Df: degrees of freedom; MS: mean square.

** $p < 0.001$.

* $p < 0.5$.

Each parameter had different effects on the mean number of alleles (A), the inbreeding coefficient (F_{IS}), and the proportion of extinct populations. The parameters with the strongest effect were maximum age that a tree can reach, germination rate, selection pressure value, and average number of seeds produced by each tree on the number of alleles (Table 2).

4.2. Model application

Complete output files from the simulated restoration experiment are available at Github (https://github.com/sujiips/Restoration_PVA). Both initial population size and initial genetic composition affected population viability and long-term maintenance of genetic diversity (Table 3). As expected, larger populations resulted in smaller effects of genetic drift, measured here as loss of total number of alleles (Fig. 2a). In populations from small areas, all simulated scenarios lead to allele loss and an increase in inbreeding coefficient.

Regarding genetic diversity, if planted saplings were produced from seeds of a single isolated tree, initial genetic diversity was already low and the number of alleles tended to one in all simulation conditions

(Fig. 2). An increase either in the number of mother trees or pollen donors resulted in greater initial genetic diversity, and reduced the rate allele loss, especially in larger initial populations (200 and 400 individuals). Results of simulations with five or more mother trees were similar in mid-term. In the short term (< 250 years), initial populations generated with seeds from five mother trees had a relatively reduced number of alleles. There was no difference in allelic diversity, effective number of alleles and allelic richness between 10 and 20 pollen donors (Fig. 2, Fig. S4).

Initial genetic composition had a strong effect on the inbreeding coefficient in populations from the small areas. Sampling seeds from only one isolated tree resulted in a larger increase in inbreeding and in stronger bottleneck effects than sampling seeds that were result of outcrossing (Fig. 2). Selection against overall homozygosity at the germination phase kept the inbreeding coefficient small or moderate, even with fixation of alleles at some loci (Fig. 2). The probability of extinction was greater in populations from smaller areas. Also, populations founded by seeds from only one isolated mother tree had higher proportion of extinct populations. Populations initiated with seeds from 30 seed sources also had high proportion of extinct populations,

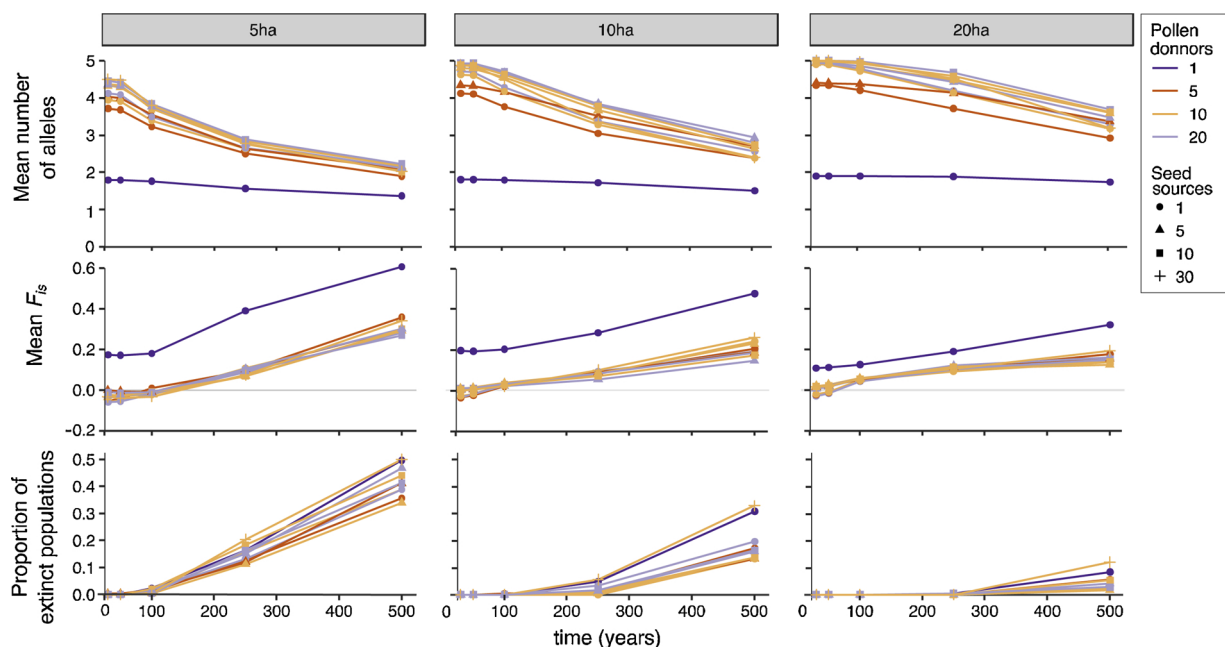


Fig. 2. Effects of initial genetic diversity and initial population size on mean number of alleles observed in each population after different time spans for each treatment; mean inbreeding coefficient (F_{IS}) observed in each population after different time spans for each treatment; and proportion of populations that became extinct after each time span for each treatment.

although the mean number of alleles and the mean F_{is} were similar to simulations with five and 10 seed sources (Fig. 2).

5. Discussion

5.1. Model development

The model we present is a tool for understanding the effects of initial genetic diversity and population size on the long-term viability of tree populations in forest restoration projects. It can be applied in studies of tree species with different characteristics, from tropical and temperate forests. This model is particularly useful when rare and threatened species are reintroduced in restoration sites to support their conservation, which relies on the potential of reintroduced populations to persist overtime (Duquesnel et al., 2017). Restoration practitioners can associate their current knowledge and new field data with the results from simulations of varied scenarios, which may comprise species with different characteristics, and a more comprehensive range of area sizes in order to guide planning and implementation of new restoration areas.

The use of spatially explicit stochastic space–time simulations has many advantages over spatial mathematical models including the capacity to include biological complexity such as stochasticity, individual-level variation, adaptive traits, as well as spatial processes (e.g., dispersal) that are difficult to reliably model using purely mathematical approaches (Epperson et al., 2010). Individual-based models are particularly useful for planning reintroduction and conservation of endangered species in restoration areas (Covington et al., 2001; Bruford et al., 2010), especially if restoration and conservation practitioners and modellers work together (Wood et al., 2015).

Habitat loss and population disruption by fragmentation are two of the main causes of species extinction (Fahrig, 1997). Forest restoration is a relevant complement to species conservation plans because it enables the increase in forest area and the reintroduction of endangered species. Nevertheless, endangered species are usually found in small population sizes and in low densities, which hampers seed sampling both for *ex-situ* and *in-situ* conservation. The model we have presented is a valuable tool for decision-making as it can be used to help design seed sampling strategies for effective and genetically informed restoration efforts. It is important to note that for many tree species, some of the parameters required by our model are not readily available; one of the unfortunate paradoxes of working with endangered, and hence rare, species. Nonetheless, data on threatened species autoecology and population structure are becoming increasingly available (Kattge et al., 2011; Sousa-Baena et al., 2014; Canhos et al., 2014; de Lima et al., 2015). Information can also be extrapolated from species with similar ecological and demographic characteristics and simulations covering the uncertainty range of the parameter can be performed.

This model accounts for overlapping generations and effects of inbreeding depression, which are present in many tree species (Angeloni et al., 2011). The effects of inbreeding depression were summarized in this model in the germination process, that is when the negative effects of inbreeding are strong and were more carefully measured (Ishida, 2006; Naito et al., 2005; Chaves et al., 2011). Although there is evidence that inbreeding may reduce survival and reproduction in different life stages (Hufford and Hamrick, 2003; Naito et al., 2005), there is a lack of studies testing the magnitude of the effect on later life stages. Further investigation into the effects of inbreeding depression on different life stages is required.

This model was designed for simulations of populations in the short and mid-term (tens of generations), because it does not account for mutation and for changes in selective pressures over time. It focuses on only one isolated population, but it is possible to evaluate effects of inter-population gene flow when a migration parameter is included. These limitations may be overcome using the results obtained from our

model as input or assumptions for other simulation programs that account for mutation, migration and natural selection in changing environments.

5.2. Model application

Restoration projects must consider both the spatial attributes of the restoration area and the initial effective population size to reduce the risk of genetic bottlenecks and to increase the probability of long-term population persistence (Allendorf et al., 2013). In general, smaller restoration areas contain smaller populations. Thus, in long narrow riparian restoration areas such as those targeted for *C. tomentosum*, high initial genetic composition (i.e., diversity) of the seed pool is necessary to maintain such diversity and assure long-term population viability. The results from our case study indicate that increasing the number of seed sources from one to five can be enough to reduce the bottleneck effect, and sampling seeds from trees that can receive pollen from at least five other trees avoids the fixation of alleles in the medium term. Given our results, for *C. tomentosum* and species with similar reproductive characteristics we recommend for forest managers to avoid sampling seeds from isolated trees and to use more than five trees from large, well preserved populations as seed sources.

In all sizes of restoration area, planting seeds from only one isolated tree results in a very strong bottleneck effect, higher extinction rate and genetic diversity loss. Increasing either the number of mother trees or the number of pollen donors increase the initial levels of genetic diversity, which is beneficial for slowing down the loss of alleles over time. Higher initial genetic diversity also resulted in lower levels of inbreeding in a mid-term.

We found that restoration projects using *C. tomentosum*, or using other species with similar life-history traits, should use seeds taken from at least ten mother trees to ensure successful restoration. Beyond this number (e.g., up to 30) we did not identify any additional long-term benefits in terms of the maintenance of genetic diversity and reduction in inbreeding. The higher proportion of extinct populations observed in simulations with seeds from 30 mother trees may be a result of the random sample of a few seeds from each tree. More simulations with defined seed sampling designs are required to explain this result. Our finding of a minimum of 10 trees is lower than current recommendations in the literature, where greater than 30 trees are recommended (Vencovsky, 1986; Basey et al., 2015). The recommendation for > 30 trees aims to conserve a minimum effective population size of 100, which was not addressed in our study. Instead, we focused on how the different levels of initial genetic diversity influenced long-term maintenance of genetic diversity in restoration populations without explicit consideration of effective population size. We show that under situations where conservation resources are limited, and management objectives involve identifying suitable trade-offs between population viability and minimizing costs, seeds taken from ten trees will be sufficient. This is a preliminary study, with only one species, and a limited number of simulated scenarios to exemplify one application of the model. The overall pattern observed here should not be used isolated to support forest restoration and management actions. More extensive studies, however, may help to improve the use of the proposed model in ecological restoration programs.

In addition, as all simulated areas showed reduction in genetic diversity and increase in inbreeding coefficient estimates, to ensure long-term sustainability, restoration programs should emphasize connectivity among newly restored patches and existing forest patches (Rudnick et al., 2012; FAO, 2015). Connectivity among forest fragments is essential to gene flow among remnant and other restoration areas increase metapopulation size and reduce the negative effects of small population sizes observed in our simulations (Frankham, 2015). Although we did not explicitly examine this question in this study, future work in this area should examine the interactions between initial

genetic diversity and spatial connectivity in fragmented landscape at different spatial scales and over varying conservation planning horizons.

6. Final remarks

Simulation approaches have been used previously in forest restoration to assess the impact of ecophysiological parameters on species resilience of forest stands (Pietsch and Hasenauer, 2002); to understand the influence of management on forest structure over time (Covington et al., 2001); and to predict habitat quality in restoration plantations (Pausas et al., 1997). Simulations are a useful tool for evaluating restoration success when empirical manipulation of the system is either too costly or limited by the rarity and conservation status of the taxa being examined. Nonetheless, integration of simulation and empirical studies will further serve to inform decision making processes and ultimately improve the probability of success and long term persistence of restoration efforts. To our knowledge, this is the first model to simulate restored population persistence using a combined population biology and population genetics approach. As a novel approach, it will require further adaptations and development to address other specific restoration projects. This model, and the approach on which it is based, have great potential to support seed collection, to guide restoration implementation, and to increase the probability of long term persistence of restoration plantations.

Author's contributions

PSS, PMAJ, MIZ and PHSB worked on the study conception and design. PSS and MEN wrote the simulation program. PSS analyzed, interpreted data and drafted the manuscript. All authors participated in critical revision of the manuscript.

Conflicts of interest

None.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.ecolmodel.2019.04.014>.

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